

Leopardus guttulus, Southern Tiger Cat

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Taxonomy

Kingdom	Phylum	Class	Order	Family
Animalia	Chordata	Mammalia	Carnivora	Felidae

Taxon Name: *Leopardus guttulus* Hensel, 1872

Common Name(s):

- English: Southern Tiger Cat, Southern Little Spotted Cat
- French: Chat tigre du sud
- Spanish: Tirica

Taxonomic Source(s):

Trigo, T.C., Schneider, A., de Oliveira, T.G., Lehugeur, L.M., Silveira, L., Freitas, T.R.O. and Eizirik, E. 2013a. Molecular data reveal complex hybridization and a cryptic species of Neotropical wild cat. *Current Biology* 23(24): 2528–2533.

Taxonomic Notes:

Taxonomy is currently under review by the IUCN SSC Cat Specialist Group's Cat Classification Task Force. Although described originally as a new species, *Felis guttula* Hensel, 1872, this species was commonly treated as a subspecies of *Leopardus tigrinus* until 2013 (Trigo *et al.* 2013a), when they demonstrated with a comprehensive molecular study that *Leopardus tigrinus* and *L. guttulus* are clearly distinct species. *L. guttulus* hybridises with *L. geoffroyi* in Rio Grande do Sul, which is the type locality. The holotype of *L. guttulus* should be investigated in order to check whether it is a hybrid. Further research is required to establish whether northwestern tigrinas comprise a third species, *L. pardinoides*, or even a fourth species, *L. onchilla*, in Costa Rica (Johnson *et al.* 1999, Trigo *et al.* 2008, Nascimento 2010). Also the distribution and morphology of *L. guttulus* need to be better defined. Until then *L. guttulus* is recognised as a monotypic species: *Leopardus guttulus* (Hensel 1872; 73).

Assessment Information

Red List Category & Criteria: Vulnerable C1 [ver 3.1](#)

Year Published: 2016

Date Assessed: May 10, 2014

Justification:

Leopardus guttulus has only recently been acknowledged as a valid species, separate from the former *Leopardus tigrinus*, due to their genetic uniqueness and differences (Trigo *et al.* 2013a). Its current northern limits of geographic distribution are still unclear, but show a large extent of occurrence of ca. 3,287,075 km². However, its area of occupancy (AOO) is rather limited and within the threatened Cerrado and Atlantic Forest domains. Considering all available natural habitat in its entire range, the maximum AOO possible would be of only 473,254 km². Additionally, given the level of fragmentation and the very small size of some patches, it is very unlikely that the Southern Tiger Cat would be present in some parts of this remaining area. Besides, no single species occupies all natural areas within its geographic range. Therefore its actual AOO would be much lower than the estimates presented here.

The species is almost always found at low/very low population densities (0.01–0.05/km² or lower), especially in protected areas, given the negative impact it suffers from Ocelots ('ocelot effect' – Oliveira *et al.* 2010, Oliveira 2011). Thus, most of the population probably occurs outside protected areas, in places under severe pressure from land development/habitat conversion. A population decline in the past few years has also been detected in Santa Catarina state in Brazil, the area considered the most important for its conservation, the estimated density dropped from 0.25/km² to 0.13/km² by almost 50% from 2005 to 2013/14, probably as a consequence of prey reduction (Oliveira *et al.* 2008, 2013, Kasper *et al.* 2016, M.A. Tortato pers. comm.). If this decline continues, a population reduction of 80% would be expected in three generations (15 years) in this area. In another area the estimated density dropped by over 50% from 0.2/km² to <0.1/km² from 2008 to 2015, resulting in a population reduction of over 7%/year. If this decline continues, the population would disappear in 3 generations in this area. Additionally, the existing global population is still suffering from habitat loss and fragmentation, which in the case of the Cerrado biome is undergoing high rates of loss (Klink and Machado 2005, MMA/IBAMA 2011). Other threats include hunting due to conflicts with rural owners, competition and diseases spread by domestic dogs, indiscriminate use of rodent poisoning, and road kills (Oliveira *et al.* 2008, 2013; Baréa and Leite 2012, R. Leite-Pitman and T. Trigo pers. comm.). It is also very important to consider the existence of an extensive hybridization zone between *L. guttulus* and *Leopardus geoffroyi* on the geographic limits of their distribution that may be affecting the genetic uniqueness of this species, as 38.2% of the southern tiger cat specimens there are introgressed from the latter (Trigo *et al.* 2008, 2013a).

A declining global population of about 6,047 mature individuals of *L. guttulus* was estimated, given its expected maximum area of occupancy (AOO) and typical lower population densities (Oliveira 2011, Oliveira *et al.* 2010, 2013). Given the likely overestimation of the AOO, using the lower known density estimate values is warranted to compensate and avoid an unrealistic population overestimation. Based on the observed continuing decline in mature individuals of 50% and more in two important conservation areas for *Leopardus guttulus* a continuing decline of over 10% over 15 years (= three generations) can be estimated. Due to its low estimated population size of less than 10,000 mature individuals and the high observed population reductions, the species may even qualify for Endangered in the near future.

Geographic Range

Range Description:

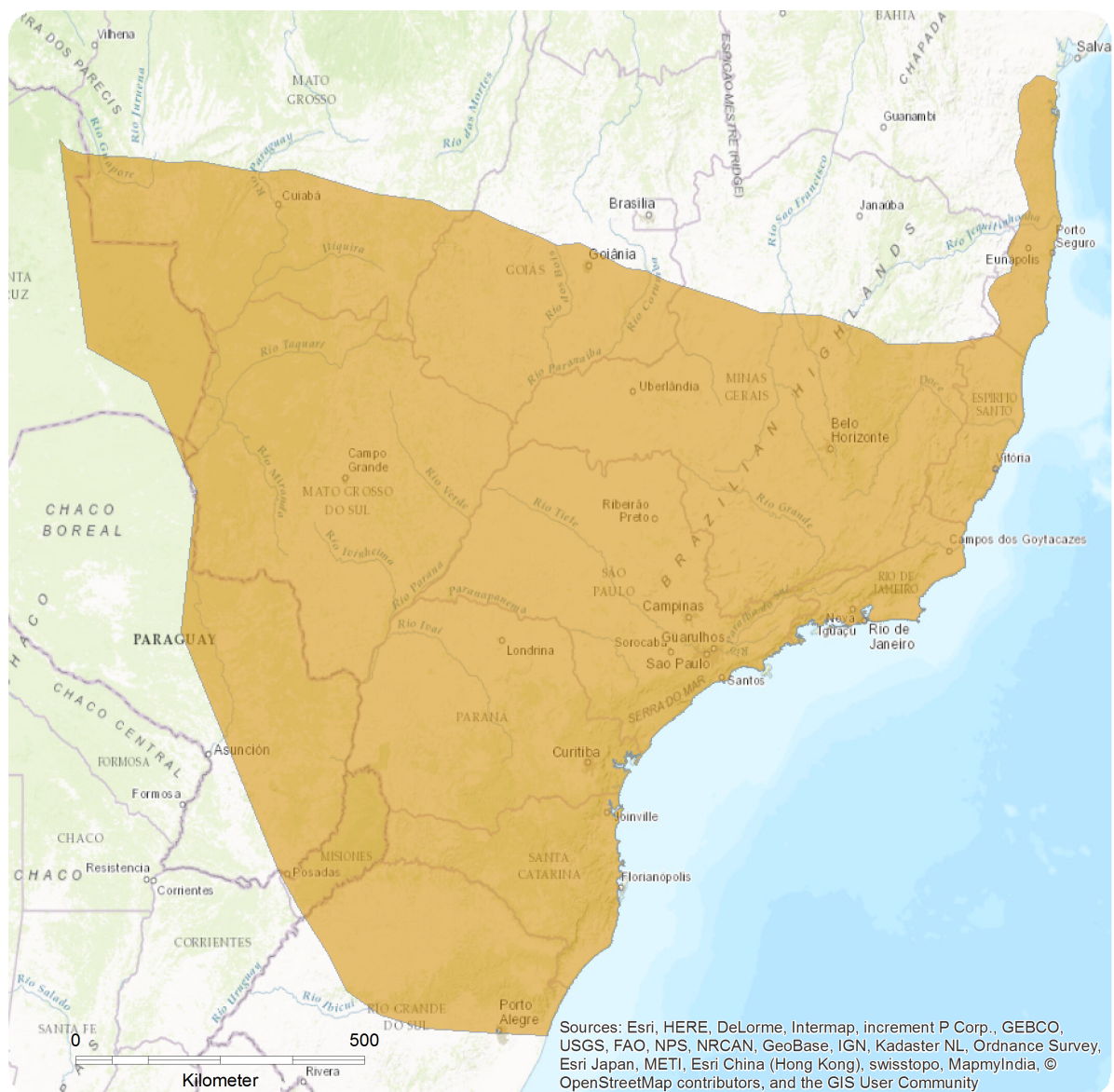
The Southern Tiger Cat ranges from Central to southern Brazil, eastern Paraguay and northeastern Argentina in the Provinces of Misiones and Corrientes. The species is absent from the Paraguayan Chaco. The northern limits of its geographic range are still unclear. It reaches Central Brazil in the states of Minas Gerais, Goiás, the border area of the Pantanal of Mato Grosso do Sul state, and the Atlantic Forest area of eastern Bahia in the northeast region (T. de Oliveira pers. comm.). Whether it overlaps with *Leopardus tigrinus* and to what extent is still unclear. Known altitudinal range of the Southern Tiger Cat is below 2,000 m (Oliveira *et al.* 2008).

Country Occurrence:

Native: Argentina (Corrientes, Misiones); Brazil (Bahia, Espírito Santo, Goiás, Mato Grosso do Sul, Minas Gerais, Paraná, Rio de Janeiro, Rio Grande do Sul, Santa Catarina, São Paulo); Paraguay

Distribution Map

Leopardus guttulus



Range

Extant (resident)

Compiled by:

IUCN (International Union for Conservation of Nature)



The boundaries and names shown and the designations used on this map do not imply any official endorsement, acceptance or opinion by IUCN.



Population

Leopardus guttulus population densities vary, but tend to be low throughout most of its range especially in regard to what would be expected of such a small cat species. Given its body size, expected density would be 91/100 km². Population densities of *L. guttulus* vary from one to 25/100 km². It is generally around 1-5/100 km² (or even lower) in most of its area of occupancy but can get to 13-25/100 km² in a few isolated high density areas (Oliveira 2011, Oliveira *et al.* 2013). Higher densities are attained only where Ocelots are absent or very rare. Where Ocelots are common or attain densities of 10/100 km² or higher, the Southern Tiger Cat is extremely rare and is expected to reach numbers much lower than 1-5/100 km². Thus, the Southern Tiger Cat is negatively impacted by Ocelots by interspecific killing due to potential for competition in what is known as the “Ocelot effect” (Oliveira *et al.* 2010, Oliveira 2011).

The state of Santa Catarina in southern Brazil has been considered the most important area for *Leopardus guttulus* conservation and thus to harbour the best and likely most viable population. This is due to the extent of the state’s remaining Atlantic Forest, level of connectivity and the cat’s presence in reasonable numbers in local Conservation Units due in most part to the absence of Ocelots (Oliveira *et al.* 2008). Nevertheless, previous high densities found in some areas there (25/100 km²) seemed to have dropped to 13/100 km² or 7/100 km² in the past few years, likely as a consequence of natural rodent reduction (Oliveira *et al.* 2013, M. Tortato pers. comm.). Therefore, expected higher densities of this species might be lower than previously thought. Considering all remaining area of natural habitat there and the species densities recorded, population size in Santa Catarina is expected to be around 1,500–2,800 (more likely towards the lower limit). Elsewhere, population reductions of *ca.* 67% have been observed due to an increase in number of Ocelots (Oliveira 2011, Marques 2013). At other areas, reductions from 20-24/100 km² to 8/100km² have also been noted (Oliveira 2011, Kasper *et al.* 2015), but not as a consequence of the Ocelot effect.

Likely as a consequence of the Ocelot effect, *L. guttulus* is found mostly outside protected areas of two of South America’s most threatened biomes, the highly fragmented Atlantic Forest, whose natural cover remaining is around 8% and the savannahs, whose remaining area is <40%. In this way, conservation efforts should focus outside protected areas.

A declining global population of about 6,047 mature individuals of *L. guttulus* was estimated, given its expected maximum area of occupancy (AOO) and typical lower population densities (Oliveira 2011; Oliveira *et al.* 2010, 2013). Given the likely overestimation of the AOO, using the lower known density estimate values is warranted to compensate and avoid an unrealistic population overestimation. A future decline of 10–30% over the next 15-18 years (= three generations) is projected due to declining AOO (habitat loss) and habitat quality. Even though the Southern Tiger Cat is found in several protected areas, no conservation units within the species’ range in Brazil or elsewhere are expected to house 500 individuals, most far below 200 (Oliveira *et al.* 2008).

Current Population Trend: Decreasing

Habitat and Ecology (see Appendix for additional information)

The Southern Tiger Cat inhabits a broad variety of habitats. These range from dense tropical and subtropical rainforests, deciduous/semideciduous, and mixed pine forests, to the open savannahs, and beach vegetation, both pristine and disturbed (Tortato and Oliveira 2005; Oliveira *et al.* 2008, 2013;

Oliveira 2011). In the Pantanal (wet/swampy savannah) it is extremely rare and has been recorded only in the dry savannahs, not in the marshy areas. Although the Southern Tiger Cat can also inhabit disturbed formations, its occurrence near agricultural fields is, nevertheless, limited by the presence of natural cover. Thus, restricted to mosaics of forest or savannah and small-scale agriculture. In these areas, both telemetry information and scat analysis indicates that it uses the natural formations and does not venture deep into the agricultural areas *per se*, but only on its borders, taking advantage of the high rodent densities (Facure-Giaretta 2002, Oliveira *et al.* 2010). The species is mostly found in lowland areas, reaching up to 2,000 m above sea level.

Litter size is usually one, sometimes two (Oliveira and Cassaro 2005). Diel activity pattern is predominantly nocturno-crepuscular, but with considerable amount of daytime activity. This activity during any time of the day is suggested to be a strategy to avoid predation by the larger sympatric Ocelot (Oliveira *et al.* 2010). Prey base consists mostly of small mammals, birds and lizards, with average prey size at <100 g, but does include larger sized prey (>1 kg) (Facure-Giaretta 2002, Oliveira *et al.* 2008, Tortato 2009, Trigo *et al.* 2013b). The limited information on home ranges shows that it is 2.5 times larger than they would be expected, based on the cat's body size. This may be another consequence to avoid other larger and potential felid predators. Known home ranges for the Southern Tiger Cat range from 2 to 25 km² (Oliveira *et al.* 2010).

Systems: Terrestrial

Use and Trade

See under Threats.

Threats (see Appendix for additional information)

The Southern Tiger Cat was heavily exploited for the fur trade decades ago, following the decline in Ocelot trade. Although international trade ceased, there is still some localized illegal hunting, usually for the domestic market. A population decline in the past few years has been detected in Santa Catarina state in Brazil, the area considered the most important for its conservation, where the density estimates dropped by more than 50%, probably as a consequence of prey reduction (Oliveira *et al.* 2008, 2013; M.A. Tortato pers. comm.). Additionally, the existing global population is still suffering from habitat loss and fragmentation, which in the case of the Cerrado biome is undergoing high rates of loss (Klink and Machado 2005, MMA/IBAMA 2011). Other threats include hunting due to conflicts with rural owners, competition and diseases spread by domestic dogs, indiscriminate use of rodent poisoning, and road kills (Oliveira *et al.* 2008, 2013; Baréa and Leite 2012; R. Leite-Pitman and T. Trigo pers. comms.). It is also very important to consider the existence of an extensive hybridization zone between *L. guttulus* and *Leopardus geoffroyi* on the geographic limits of their distribution that may be affecting the genetic uniqueness of this species, as 38.2% of the Southern Tiger Cat specimens there are introgressed from the latter (Trigo *et al.* 2008, Trigo *et al.* 2013a).

Conservation Actions (see Appendix for additional information)

Leopardus guttulus is not formally included on CITES Appendix I as a separate species yet (but would be automatically included as *L. tigrinus*). Prior to its separation from *L. tigrinus*, its commercial use was legally forbidden. Technically, as all Felidae not listed on Appendix I are in Appendix II, it would automatically be under Appendix II. Hunting of the species is prohibited in Argentina, Brazil, and

Paraguay (Nowell and Jackson 1996). This species is Vulnerable in Argentina – still under *L. tigrinus* (Diaz and Ojeda 2000) and Brazil (MMA 2014). Populations in protected areas are expected to be very low, probably because of the impact of the higher Ocelot densities and, as such, conservation efforts will have to focus outside protected areas (Oliveira *et al.* 2008, Oliveira 2011). These areas are, in turn, under intense pressure from impacts, either of fragmentation and/or area loss. Further studies are required on the species' ecology, demographics, natural history, and threats.

Credits

Assessor(s): de Oliveira, T., Trigo, T., Tortato, M., Paviolo, A. & Bianchi, R. and Leite-Pitman, M.R.P.

Reviewer(s): Nowell, K., Hunter, L., Schipper, J., Breitenmoser-Wursten, C. & Lanz, T. and Breitenmoser, U.

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Appendix

Habitats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Habitat	Season	Suitability	Major Importance?
1. Forest -> 1.5. Forest - Subtropical/Tropical Dry	-	Suitable	Yes
1. Forest -> 1.6. Forest - Subtropical/Tropical Moist Lowland	-	Suitable	Yes
1. Forest -> 1.7. Forest - Subtropical/Tropical Mangrove Vegetation Above High Tide Level	-	Unknown	-
1. Forest -> 1.8. Forest - Subtropical/Tropical Swamp	-	Unknown	-
1. Forest -> 1.9. Forest - Subtropical/Tropical Moist Montane	-	Suitable	Yes
2. Savanna -> 2.1. Savanna - Dry	-	Suitable	Yes

Threats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Threat	Timing	Scope	Severity	Impact Score
1. Residential & commercial development -> 1.1. Housing & urban areas	Ongoing	Majority (50-90%)	Rapid declines	Medium impact: 7
	Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation		
1. Residential & commercial development -> 1.2. Commercial & industrial areas	Ongoing	Majority (50-90%)	Rapid declines	Medium impact: 7
	Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation		
2. Agriculture & aquaculture -> 2.2. Wood & pulp plantations -> 2.2.2. Agro-industry plantations	Ongoing	Majority (50-90%)	Very rapid declines	High impact: 8
	Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation		
2. Agriculture & aquaculture -> 2.3. Livestock farming & ranching -> 2.3.3. Agro-industry grazing, ranching or farming	Ongoing	Majority (50-90%)	Rapid declines	Medium impact: 7
	Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation		
4. Transportation & service corridors -> 4.1. Roads & railroads	Ongoing	Minority (50%)	-	-
	Stresses:	2. Species Stresses -> 2.1. Species mortality		
5. Biological resource use -> 5.1. Hunting & trapping terrestrial animals -> 5.1.3. Persecution/control	Ongoing	Minority (50%)	Causing/could cause fluctuations	Low impact: 5
	Stresses:	2. Species Stresses -> 2.1. Species mortality		

7. Natural system modifications -> 7.1. Fire & fire suppression -> 7.1.1. Increase in fire frequency/intensity	Ongoing	Minority (50%)	Causing/could cause fluctuations	Low impact: 5
	Stresses:	1. Ecosystem stresses -> 1.2. Ecosystem degradation		
7. Natural system modifications -> 7.3. Other ecosystem modifications	Ongoing	-	-	-
	Stresses:	1. Ecosystem stresses -> 1.2. Ecosystem degradation		
8. Invasive and other problematic species, genes & diseases -> 8.1. Invasive non-native/alien species/diseases -> 8.1.1. Unspecified species	Ongoing	Majority (50-90%)	Causing/could cause fluctuations	Medium impact: 6
	Stresses:	2. Species Stresses -> 2.3. Indirect species effects -> 2.3.1. Hybridisation		
8. Invasive and other problematic species, genes & diseases -> 8.3. Introduced genetic material	Ongoing	Unknown	-	-
	Stresses:	2. Species Stresses -> 2.3. Indirect species effects -> 2.3.1. Hybridisation		
8. Invasive and other problematic species, genes & diseases -> 8.4. Problematic species/disease of unknown origin -> 8.4.1. Unspecified species	Ongoing	-	-	-
	Stresses:	2. Species Stresses -> 2.1. Species mortality		
8. Invasive and other problematic species, genes & diseases -> 8.6. Diseases of unknown cause	Ongoing	-	-	-
	Stresses:	2. Species Stresses -> 2.1. Species mortality		

Conservation Actions in Place

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Conservation Actions in Place
In-Place Research, Monitoring and Planning
Action Recovery plan: No
In-Place Land/Water Protection and Management
Conservation sites identified: Yes, over part of range
Occur in at least one PA: Yes
Percentage of population protected by PAs (0-100): 1-10
Area based regional management plan: Yes
Invasive species control or prevention: No
In-Place Species Management
Harvest management plan: No
Successfully reintroduced or introduced benignly: No
Subject to ex-situ conservation: No
In-Place Education

Conservation Actions in Place
Subject to recent education and awareness programmes: Yes
Included in international legislation: Unknown
Subject to any international management/trade controls: Unknown

Conservation Actions Needed

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Conservation Actions Needed
1. Land/water protection -> 1.1. Site/area protection
1. Land/water protection -> 1.2. Resource & habitat protection
2. Land/water management -> 2.2. Invasive/problematic species control
2. Land/water management -> 2.3. Habitat & natural process restoration
5. Law & policy -> 5.2. Policies and regulations

Research Needed

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Research Needed
1. Research -> 1.1. Taxonomy
1. Research -> 1.2. Population size, distribution & trends
1. Research -> 1.3. Life history & ecology
1. Research -> 1.5. Threats
2. Conservation Planning -> 2.1. Species Action/Recovery Plan
3. Monitoring -> 3.1. Population trends
3. Monitoring -> 3.4. Habitat trends

Additional Data Fields

Distribution
Estimated area of occupancy (AOO) (km ²): 473254
Continuing decline in area of occupancy (AOO): Yes
Estimated extent of occurrence (EOO) (km ²): 3287075
Continuing decline in extent of occurrence (EOO): Yes
Extreme fluctuations in extent of occurrence (EOO): No

Distribution
Continuing decline in number of locations: Unknown
Extreme fluctuations in the number of locations: Unknown
Lower elevation limit (m): 0
Upper elevation limit (m): 2000
Population
Number of mature individuals: 6047
Continuing decline of mature individuals: Yes
Extreme fluctuations: No
Population severely fragmented: Yes
Continuing decline in subpopulations: Unknown
Extreme fluctuations in subpopulations: Unknown
All individuals in one subpopulation: Unknown
Habitats and Ecology
Generation Length (years): 4-5
Movement patterns: Not a Migrant

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